MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Physical Quantities & Measuring Techniques

1.1

STUDENT NAME

0315 0805775

PHYSICS WITH MOIZ DAWOODI

O'Levels 1.1

Physical Quantities & Measurement Techniques

MOIZ DAWOODI
O/A LEVEL PHYSICS

SYLLABUS OUTLINE:

1.1	Physical quantities and measurement techniques
1	Describe how to measure a variety of lengths with appropriate precision using tapes, rulers and micrometers (including reading the scale on an analogue micrometer)
2	Describe how to use a measuring cylinder to measure the volume of a liquid and to determine the volume of a solid by displacement
3	Describe how to measure a variety of time intervals using clocks and digital timers
4	Determine an average value for a small distance and for a short interval of time by measuring multiples (including the period of oscillation of a pendulum)
5	Understand that a scalar quantity has magnitude (size) only and that a vector quantity has magnitude and direction
6	Know that the following quantities are scalars: distance, speed, time, mass, energy and temperature
7	Know that the following quantities are vectors: displacement, force, weight, velocity, acceleration, momentum, electric field strength and gravitational field strength
8	Determine, by calculation or graphically, the resultant of two vectors at right angles

QUANTITIES:

In Physics, anything which is being talked about are called quantities.

Quantities
- Physical
- Non-Physical

PHYSICAL QUANTITIES:

"Quantities which can be measured are called Physical Quantities."

They are always represented by units.

E.g: Length, mass, time etc.

session21.22smd@gmail.com 0315-0805775 Physics with Moiz Dawoodi

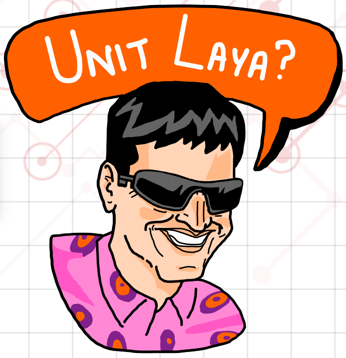
NON-PHYSICAL QUANTITIES:

"Quantities which can't be measured are called non-Physical Quantities."

They are not represented by units.

E.g: Like, sorrow, hate etc.

Physical Quantities
- Fundamental Quantities
- Derived Quantities



FUNDAMENTAL QUANTITIES:

QUANTITIES	SYMBOLS	S.I UNIT	SYMBOLS
Length	L / l	Meter	m
Mass	M / m	Kilogram	kg
Time	T / t	Second	s
Electric Current	I	Ampere	A
Temperature	T	Kelvin	K

There are total of seven Fundamental Quantities. Amount of substance & Luminous Intensity are not the part of syllabus.

Quantities that are written in Capital letters are in the memory of Scientists.

Derived Quantities:

QUANTITIES	SYMBOLS	S.I UNIT	SYMBOLS
Speed / Velocity	v	Meter/second	m/s
Acceleration	a	Meter/second^2	m/s^2
Force	F	Newton	N

Standard Form:

"A method which is used to mention a quantity which is either very small <u>or</u> very large."

EXAMPLES:

• The average diameter of gas molecule is 0.000 000 0003 m.

$\cdot \longrightarrow 10^{-x}$

$$0.\underbrace{000}_{3}\,\underbrace{000}_{6}\,\underbrace{000}_{9}\,3 = 0.3 \times 10^{-9}\ m = 0.3\ nm$$

• The total mass of the trailer is 1000 000 000 g.

$10^{+x} \longleftarrow \cdot$

$$1\,\underbrace{000}_{9}\,\underbrace{000}_{6}\,\underbrace{000}_{3}.\ g = 1.0 \times 10^{9}\ g = 1.0\ Gg$$

Exercise:

• $2\,\underbrace{000}_{9}\,\underbrace{000}_{6}\,\underbrace{000}_{3}.\ s = 2.0 \times 10^{9}\ s = 2.0\ Gs$

• $0.\underbrace{00}_{2}\,\underbrace{0004}_{6}\ m = 4.0 \times 10^{-6}\ m = 4.0\ \mu s$

• $12\,\underbrace{000}_{12}\,\underbrace{000}_{9}\,\underbrace{000}_{6}\,\underbrace{000}_{3}.\ g = 12.0 \times 10^{12}\ g = 12.0\ Tg$

• $0.\underbrace{00}_{2}\,\underbrace{000}_{5}\,\underbrace{0004}_{9}\ N = 4.0 \times 10^{-9}\ N = 4.0\ nN$

Prefix:

Words that comes before any unit & changes its value is called Prefix.

PREFIX	SYMBOLS	10^x	NUMERIC VALUE
Tera	T	10^{12}	1000 000 000 000
Giga	G	10^{9}	1000 000 000
Mega	M	10^{6}	1000 000
Kilo	k	10^{3}	1000
Hecta	H	10^{2}	100
De ca	D	10^{1}	10
Zero	-	-	0
Deci	d	10^{-1}	0.1
Centi	c	10^{-2}	0.01
Milli	m	10^{-3}	0.001
Micro	μ	10^{-6}	0.00 0001
Nano	n	10^{-9}	0.00 000 0001
Pico	p	10^{-12}	0.00 000 000 0001

UNIT CONVERSION:

LENGTH:

- 1 km = 1000 m
- 1 m = 100 cm
- 1 cm = 10 mm

EXERCISE:

- $\boxed{3}$ x km = $\boxed{3}$ x 1000 m = 3000 m
- $\boxed{10}$ x m = $\boxed{10}$ x 100 cm = 1000 cm
- $\boxed{0.2}$ x cm = $\boxed{0.2}$ x 10 mm = 2 mm
- $\boxed{4}$ x m = $\boxed{4}$ ÷ 1000 km = 0.004 km
- $\boxed{0.6}$ x cm = $\boxed{0.6}$ ÷ 100 m = 0.006 m
- $\boxed{10}$ x mm = $\boxed{10}$ ÷ 10 cm = 1 cm

Business Studies X
Big → Small *
Small → Big ÷

MASS:

- 1 kg = 1000 g
- 1 g = 1000 mg
- 1 kg = 1000000 mg

EXERCISE:

- $\boxed{3}$ x kg = $\boxed{3}$ x 1000 g = 3000 g
- $\boxed{10}$ x g = $\boxed{10}$ x 1000 mg = 10000 mg
- $\boxed{0.2}$ x kg = $\boxed{0.2}$ x 1000000 mg = 200000 mg
- $\boxed{4}$ x g = $\boxed{4}$ ÷ 1000 kg = 0.004 kg
- $\boxed{0.6}$ x mg = $\boxed{0.6}$ ÷ 1000 g = 0.006 g
- $\boxed{10}$ x mg = $\boxed{10}$ ÷ 1000000 kg = 0.00001 kg

Business Studies X
Big → Small *
Small → Big ÷

TIME:

- 1 hour = 60 minute
- 1 minute = 60 second
- 1 hour = 3600 second

EXERCISE:

- $\boxed{3}$ x h = $\boxed{3}$ x 60 minute = 180 minute
- $\boxed{10}$ x h = $\boxed{10}$ x 3600 second = 36000 second
- $\boxed{2}$ x minute = $\boxed{2}$ x 60 second = 120 second
- $\boxed{4}$ x minute = $\boxed{4}$ ÷ 60 hour = 0.066 hour
- $\boxed{0.6}$ x s = $\boxed{0.6}$ ÷ 3600 hour = 0.00016 hour
- $\boxed{10}$ x s = $\boxed{10}$ ÷ 60 minute = 0.166 minute

Business Studies X
Big → Small *
Small → Big ÷

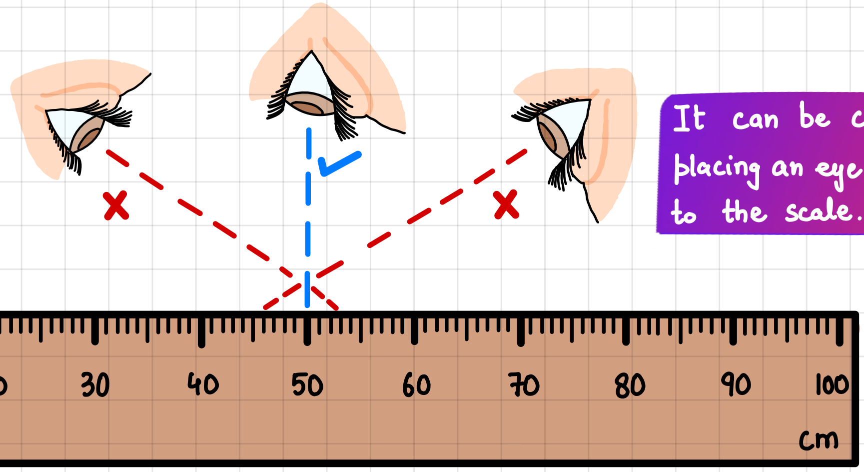
LENGTH MEASUREMENT:

INSTRUMENT	LENGTH	PRECISION
Meter Rule	Between several cm → 1m (Straight Length)	0.1 cm
Measuring Tape	Several m / Curved Length	0.1 cm
Vernier Calippers	2-10cm / Internal diameter	0.01 cm
Micrometer Screw Gauge	Less than 2cm	0.01 mm

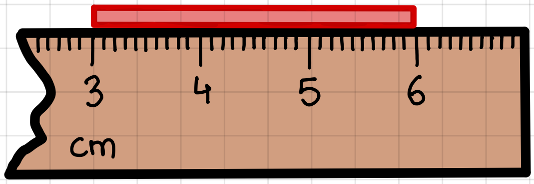
Meter Rule:

Parallax error is an error which rises due to incorrect line of sight.

It can be corrected by placing an eye perpendicular to the scale.



Due to wear & tear 0cm mark of meter rule is missing to measure length, start measuring from 1ST visible division & subtract the length from final reading.

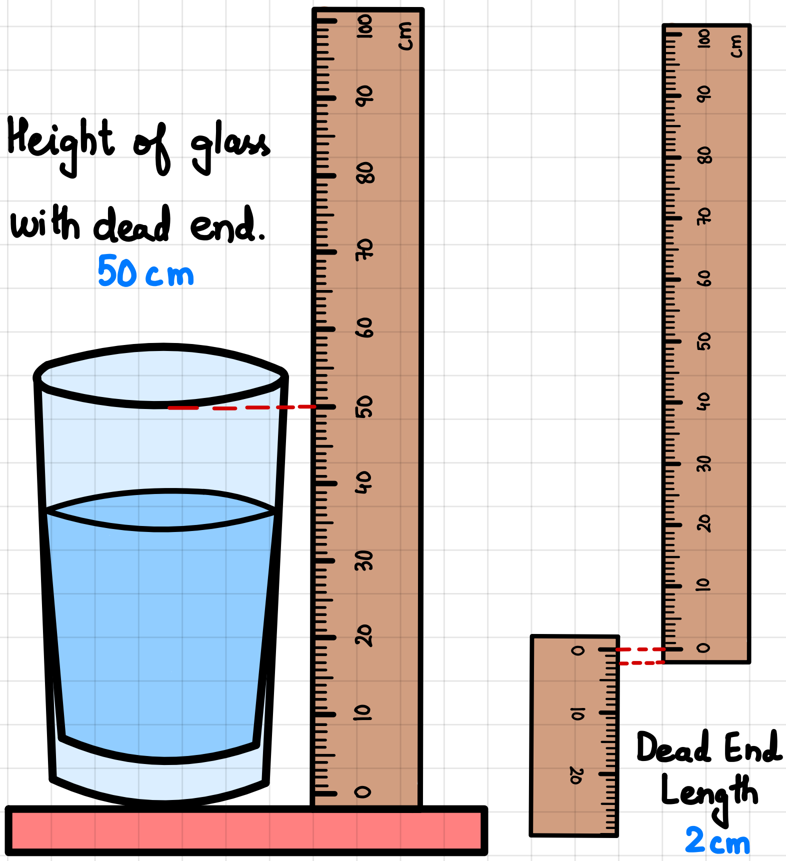


Length of bar = 6−3 = 3cm

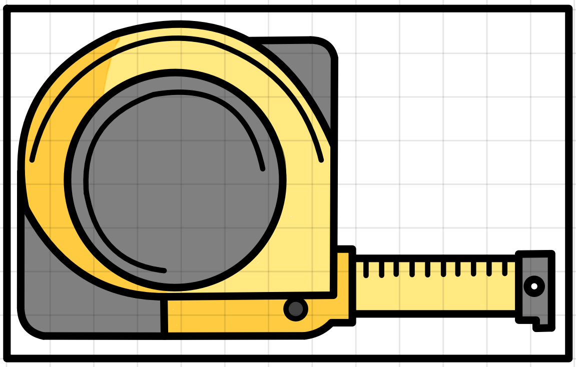
Dead End Space:

Dead end space is an unmarked length at the end of the ruler. We can find out the dead end length by using another ruler & add it to final length.

Final Height = 50+2 = 52cm

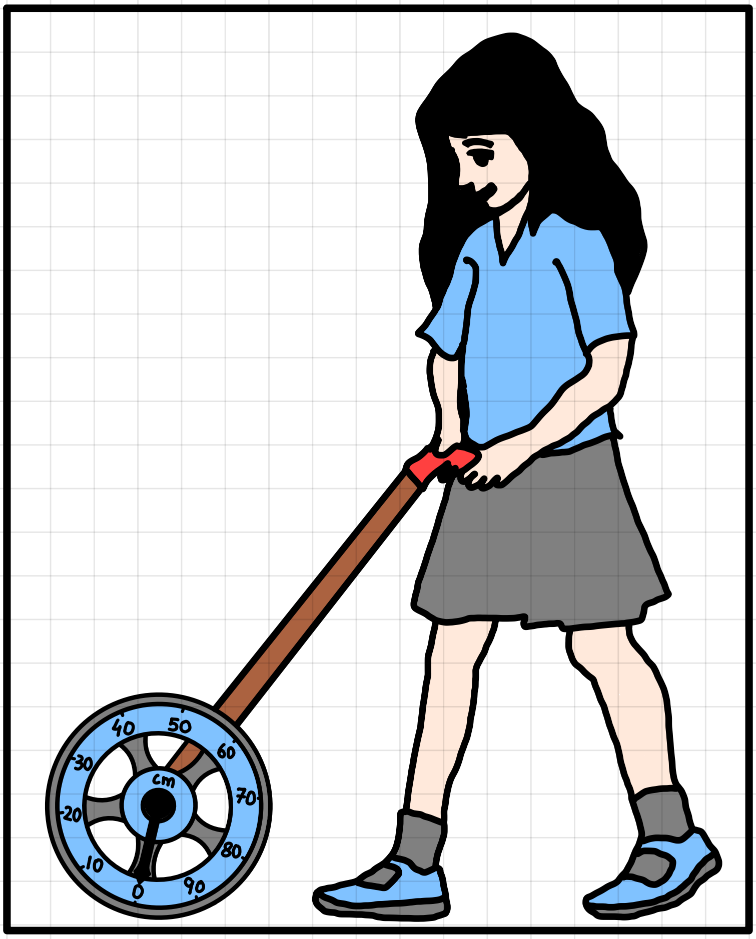


MEASURING TAPE:



Measuring tape is a flexible ruling tape which can be used to measure length above 1m to several meter. It can also measure lengths of curved path.

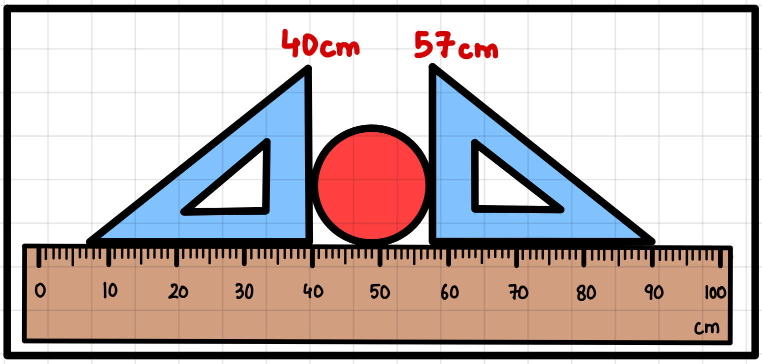
TRUNDELL WHEEL:



An instrument which is used to measure large lengths. It is used to measure on floor & pushed against the floor so wheel rotates. When one rotation completes which means 1m is covered. A tick sound is produced after every complete rotation.

To measure 70m, 70 tick sounds will be heard.

CALIPPERS:

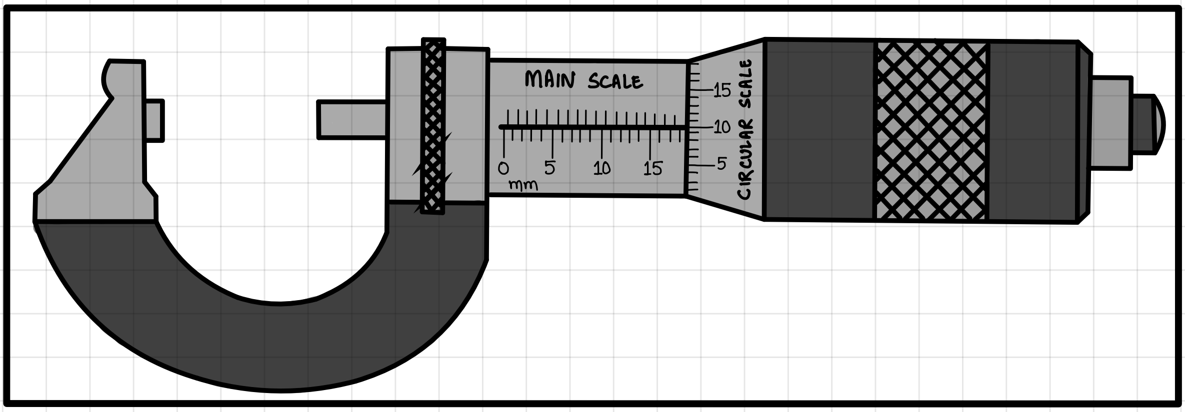


An instrument which is used to measure diameter of sphere or cylinder is called calippers.

To find diameter, subtract both readings occured at set squares.

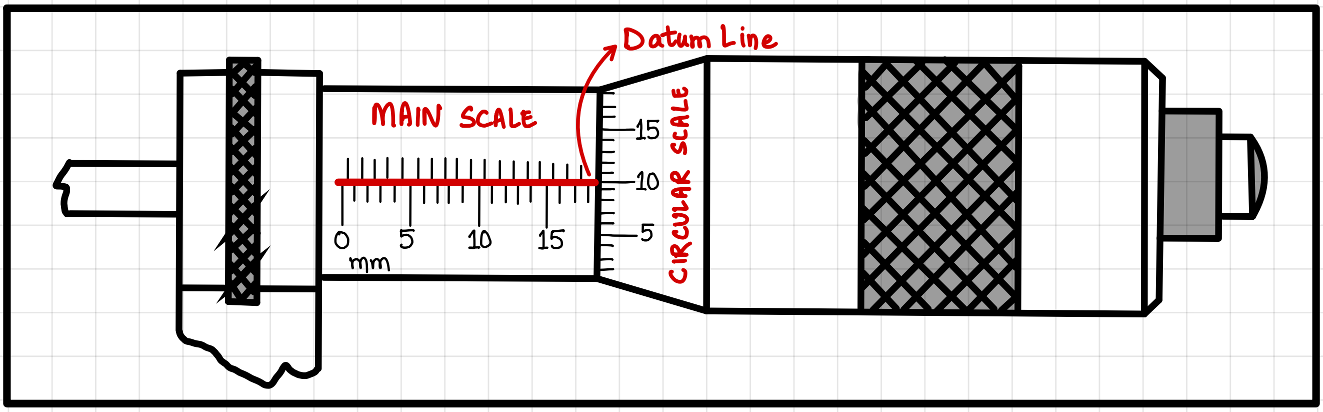
Diameter = 57 - 40 = 17cm

Micrometer Screw Gauge:



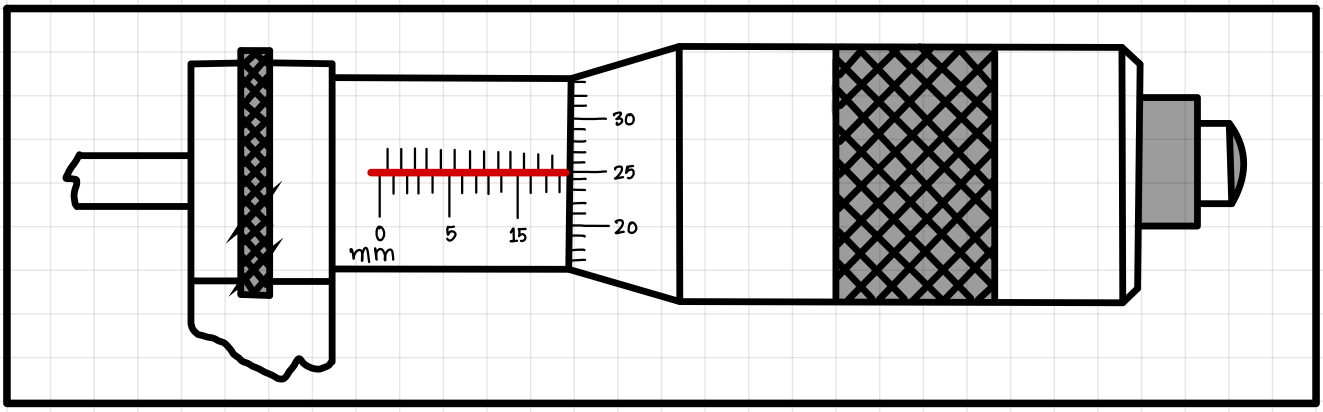
Micrometer screw gauge is a length measuring instrument which can measure length accurately less than 2cm with a precision of 0.01 mm.

① Find out the length shown by micrometer screw gauge.



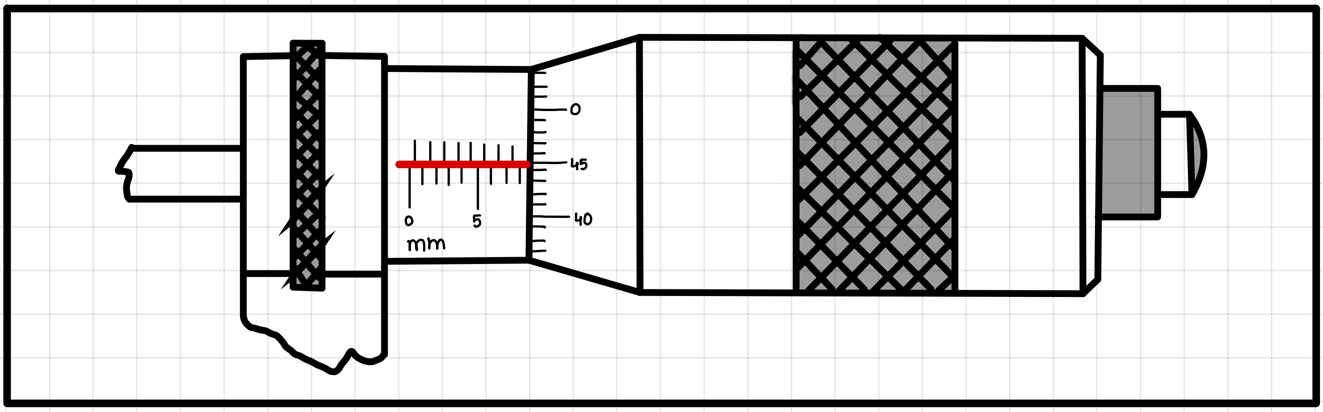
- Observe the last visible division on main scale 18mm
- Observe which division of circular scale coincides with datum line. 10th
- Multiply coincident division with precision. 10 x 0.01 mm = 0.1mm
- Add step 1 & step 3. 18+0.1 = 18.1mm

② Find out the length shown by micrometer screw gauge.



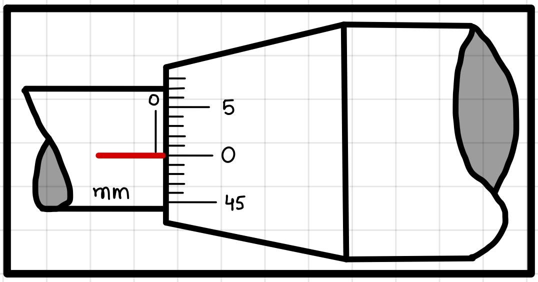
- Observe the last visible division on main scale 18mm
- Observe which division of circular scale coincides with datum line. 25^{th}
- Multiply coincident division with precision. $25 \times 0.01 = 0.25$mm
- Add step 1 & step 3. $18 + 0.25 = 18.25$mm

③ Find out the length shown by micrometer screw gauge.



- Observe the last visible division on main scale 8mm
- Observe which division of circular scale coincides with datum line. 45^{th}
- Multiply coincident division with precision. $45 \times 0.01 = 0.45$mm
- Add step 1 & step 3. $8 + 0.45 = 8.45$mm.

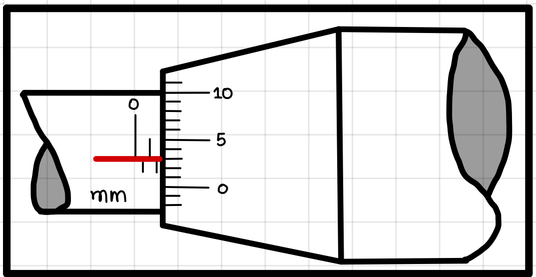
ZERO ERROR:



When jaws of micrometer screw gauge are completely closed, observe the zero of circular scale, it must coincide with datum line.

If zero of circular scale doesn't coincide with datum line, then micrometer screw gauge has a zero error.

POSITIVE ZERO ERROR:



- 1.5mm
- 3rd
- $3 \times 0.01 = 0.03$mm
- $1.5 + 0.03 = +1.53$mm

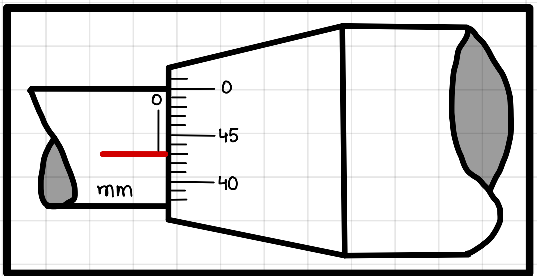
When datum line is above the zero of the circular scale, it is called positive zero error.

Zero error can be found by the same 4 steps.

+1.53 mm is absurdly added in the final reading.

To make the final reading accurate, we subtract it from final reading.

NEGATIVE ZERO ERROR:



- 0mm
- 43rd
- $43 \times 0.01 = 0.43$mm
- $0 + 0.43 = -0.43$mm

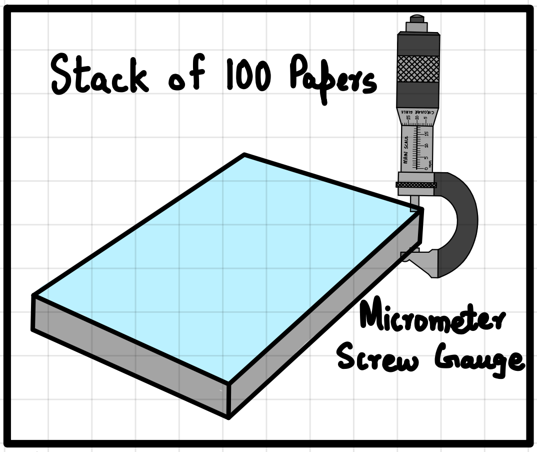
When datum line is below the zero of the circular scale, it is called negative zero error.

Zero error can be found by the same 4 steps.

-0.43 mm absurdly subtracts from the final reading.

To make the final reading accurate, we add it from final reading.

DETERMINING SMALL LENGTH:



In order to find small distance like thickness of single sheet of paper, we find thickness of 100 paper & divide it by 100 to get average thickness.

VOLUME MEASUREMENT:

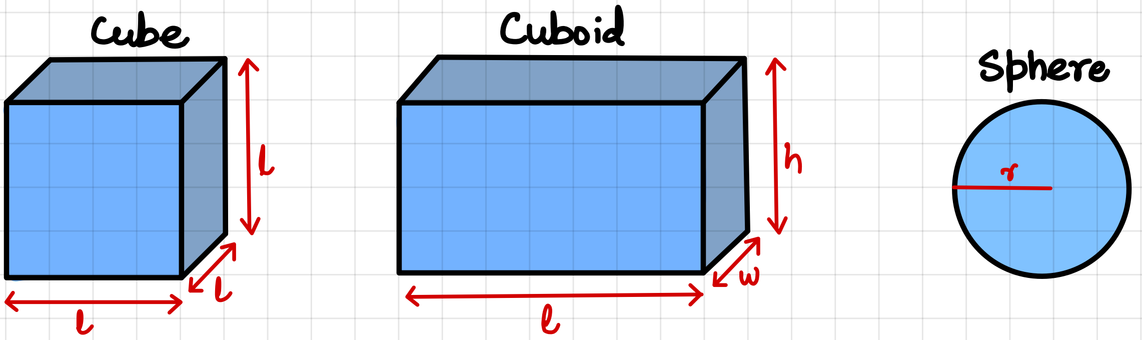
VOLUME:

"The space occupied by a body is called volume."

Its S.I unit is meter cube (m^3).

VOLUME OF REGULARLY SHAPED BODIES:

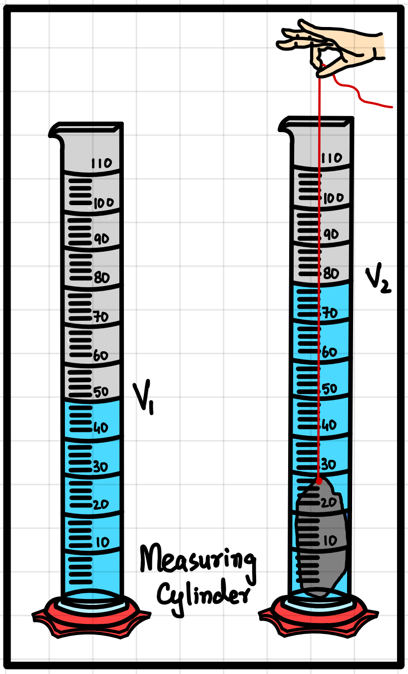
Identify the shape & apply formula of volume.



$V = L^3$ $\quad$ $V = L \times w \times h$ $\quad$ $V = \frac{4}{3}\pi r^3$

VOLUME OF IRREGULARLY SHAPED BODIES:

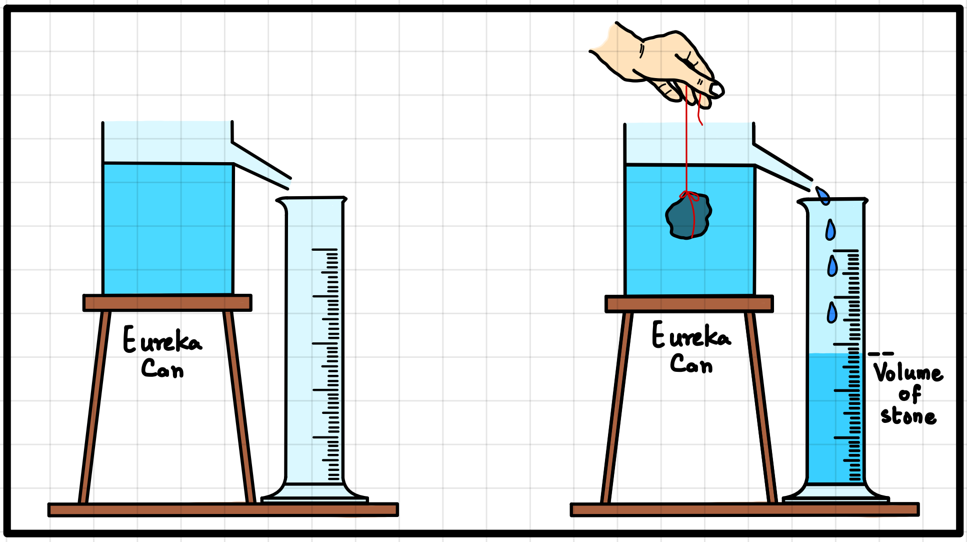
A) DISPLACEMENT METHOD:



- Pour fixed amount of water in measuring cylinder & name the volume of water as 'V_1'.
- Tie irregular stone with a thread & lower stones in measuring cylinder.
- Shake measuring cylinder gently to remove air bubbles.
- There will be new level of water. Mark this as 'V_2'.
- To find volume of stone, subtract $V = V_2 - V_1$.

B) EUREKA METHOD:

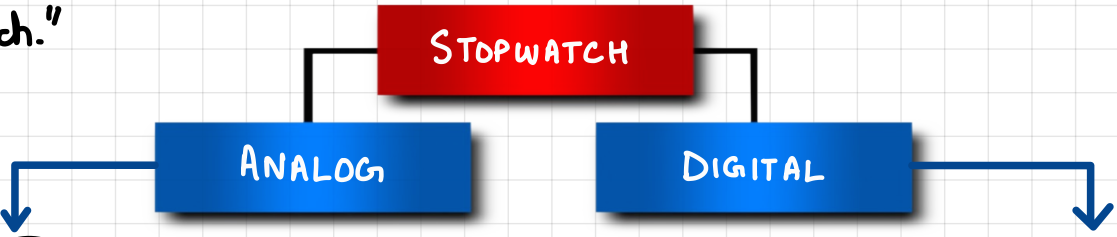
- Fill the eureka can up to nozil with water.
- Tie stone with thread & lower it into water.
- Stone displaces some amount of water in measuring cylinder which falls through nozil in measuring cylinder.
- Measure the volume of water, this volume is equivalent to volume of stone.

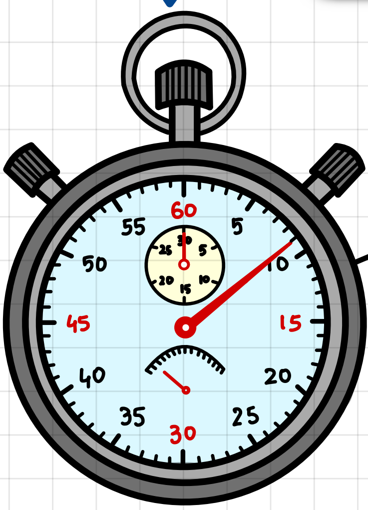


Time Measurement:

Stopwatch:

"A device which can be used to measure time intervals is called stopwatch."



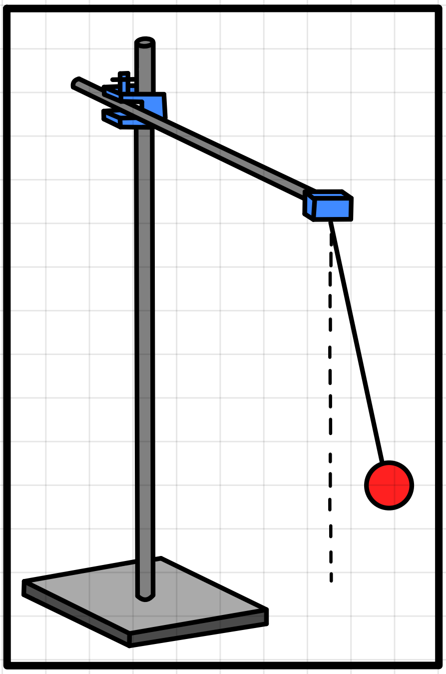


Time = 9s

Time = 3min 54s
= 180 + 54
= 234s

Simple Pendulum:

"A device made up of rigid thread tied with a mass at one end while other end is fixed with support."



OSCILLATION:

"When pendulum comes back to its original position, its called one oscillation."

TIME PERIOD:

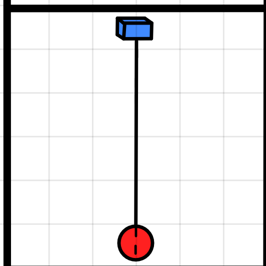
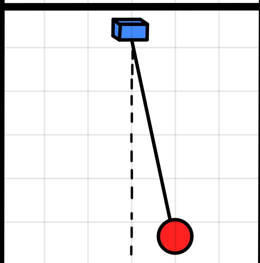
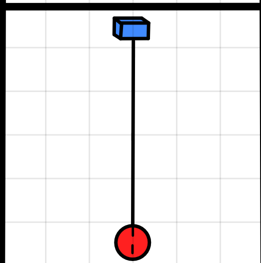
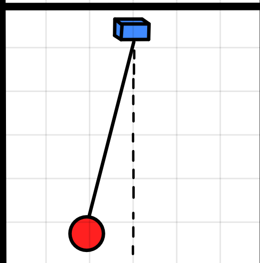
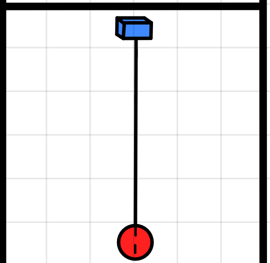
"Time required to complete one oscillation is time period."

SCAN ME
Measurement of Time Period of Simple Pendulum

SCAN ME
Simple Pendulum Simulation

MEASURE TIME PERIOD OF SIMPLE PENDULUM:

It is always suggested that measurement of time period of Simple Pendulum should start from the mean position as pendulum will always reach the mean position. To make the central position more visible, we mark central position by red colour which is called 'Fudicial Mark.'

Position 1	Position 2	Position 3	Position 4	Position 5
				
Mean	Extreme Right	Mean	Extreme Left	Mean

- To measure average time period time multiple oscillations (20) & divide total time by number of oscillations.

SCALARS:

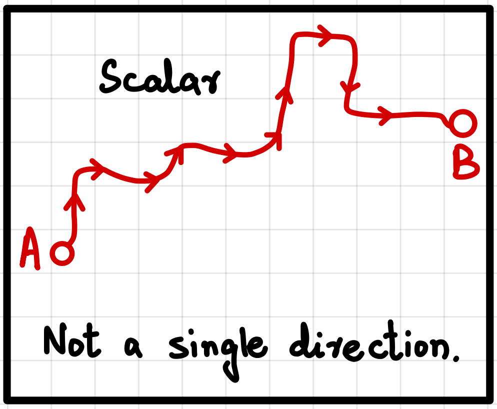
"Quantities which are specified by magnitude & unit only. They don't have specific direction."

E.g: Mass of Anas Abdul Habib is 50 kg.

Here 50 is magnitude & kg is unit.

Some of the scalar quantities are:

Distance, Speed, Time, Mass, Energy, Temperature etc.



VECTORS:

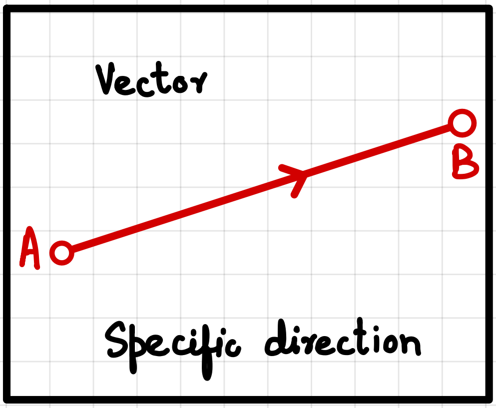
"Quantities which are specified by magnitude, unit as well as specific direction are called vectors."

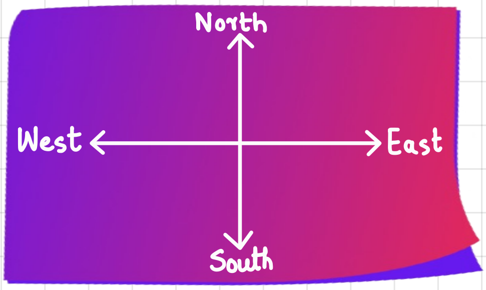
E.g: Weight of Anas Abdul Habib is 500 N downwards.

Here 500 is magnitude & N is unit & downward is direction.

Some of the vector quantities are:

Displacement, Force, Weight, Velocity, Acceleration, Momentum, Electric field strength, Gravitational Field Strength etc.





Representation Of A Vector:

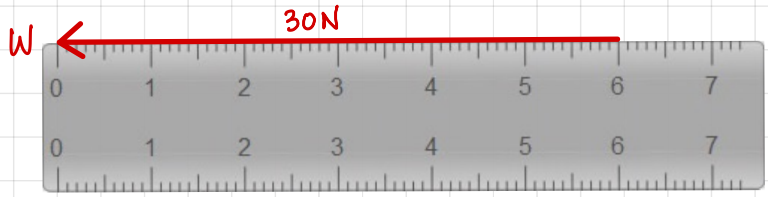
Scale shouldn't be too large or too small.
Use sharp pencil to draw vectors.

- Draw a vector of 30 N towards West.

Scale:

If 1cm = 5N
Then x = 30N

$$x = \frac{30}{5} = 6\,cm$$

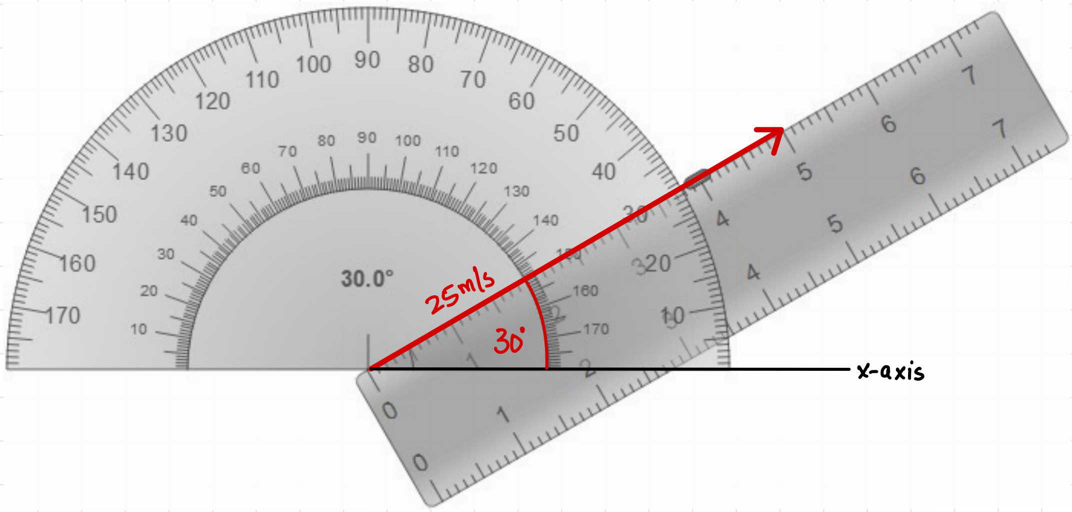


- Draw a vector of $25\,ms^{-1}$ at an angle of 30° w.r.t x-axis.

Scale:

If 1cm = 5 m/s
Then x = 25 m/s

$$x = \frac{25}{5} = 5\,cm$$



- Draw a vector of 12 Pa at N20°W.

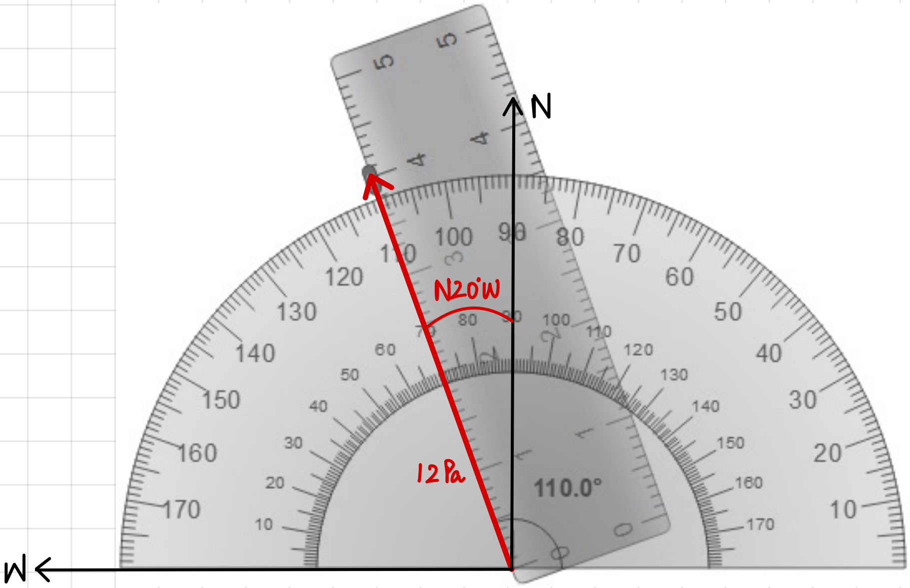
Scale:

If 1cm = 3Pa
Then x = 12 Pa

$$x = \frac{12}{3} = 4\,cm$$

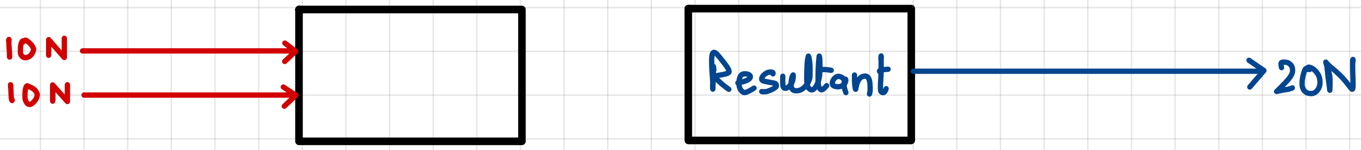
SCAN ME

Representation of a Vector



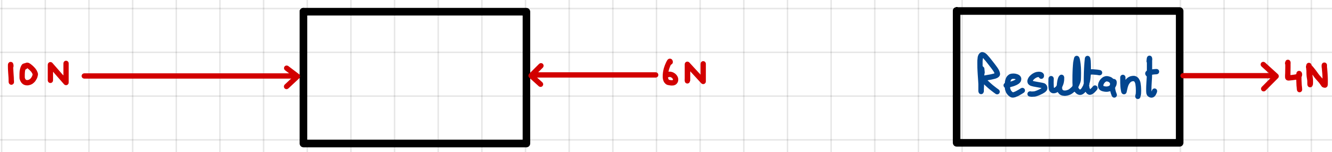
Addition Of Vectors:

A) WHEN VECTORS ARE ACTING IN SAME DIRECTION:



When vectors are acting in same direction, we add them & resultant vector is in same direction as original vectors.

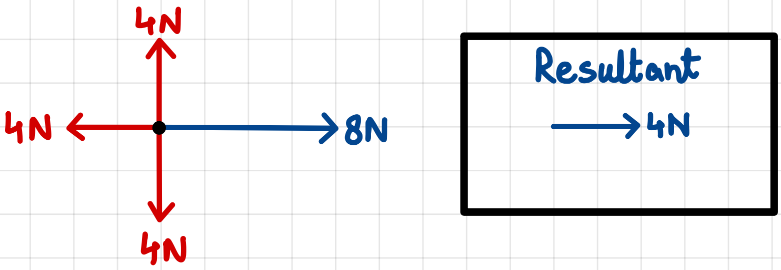
B) WHEN VECTORS ARE ACTING IN OPPOSITE DIRECTION:



When vectors are acting in opposite direction, we subtract them & resultant vector is in the direction of larger vector.

EXERCISE:

What is the direction of resultant vector?



The length of the arrow represents the magnitude of vector while arrow head represents direction

c) WHEN VECTORS ARE MAKING AN ANGLE WITH EACH OTHER:

Tip To Tail Rule:

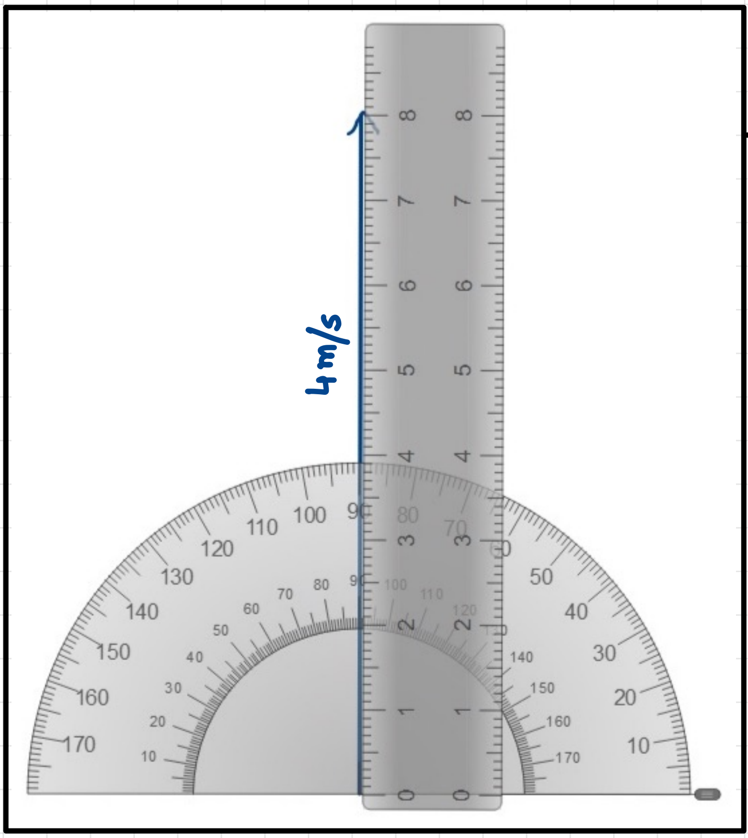
① Choose 1^{st} vector i.e. 4m/s towards north. Draw this vector according to scale.

② Draw 2^{ND} vector of 2m/s towards east by joining the tail of 2^{ND} vector with the tip of 1^{ST} vector without any change in magnitude & direction.

③ Draw resultant vector which starts from the tail of 1^{ST} vector & ends at the tip of 2^{ND} vector.

④ Measure the length of resultant vector & convert it according to scale. This is the magnitude of resultant vector.

⑤ Measure the angle of resultant vector. This is the direction of resultant vector.

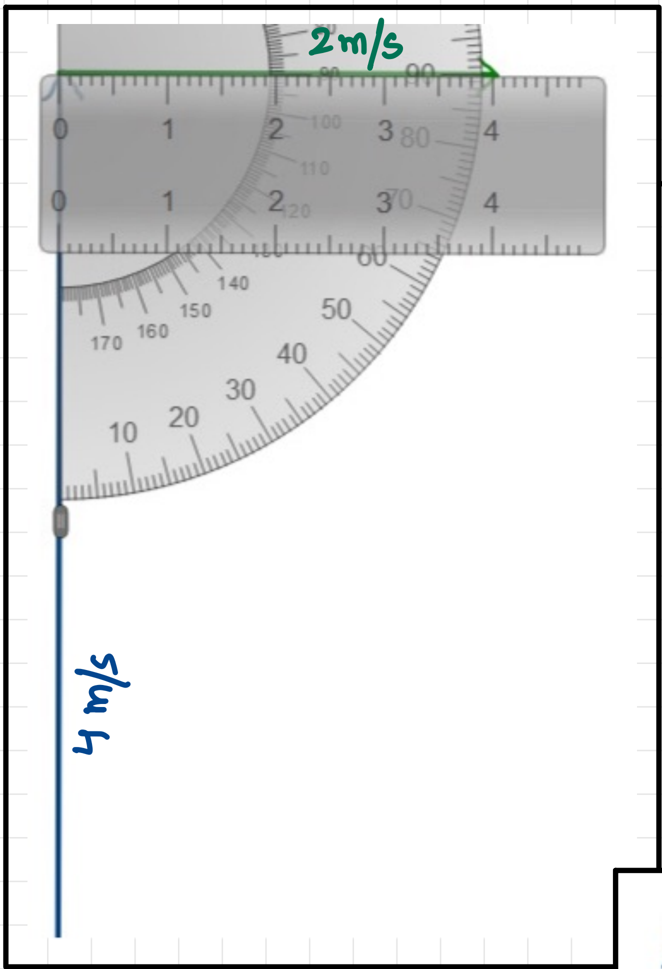


Visualize representation of step 1

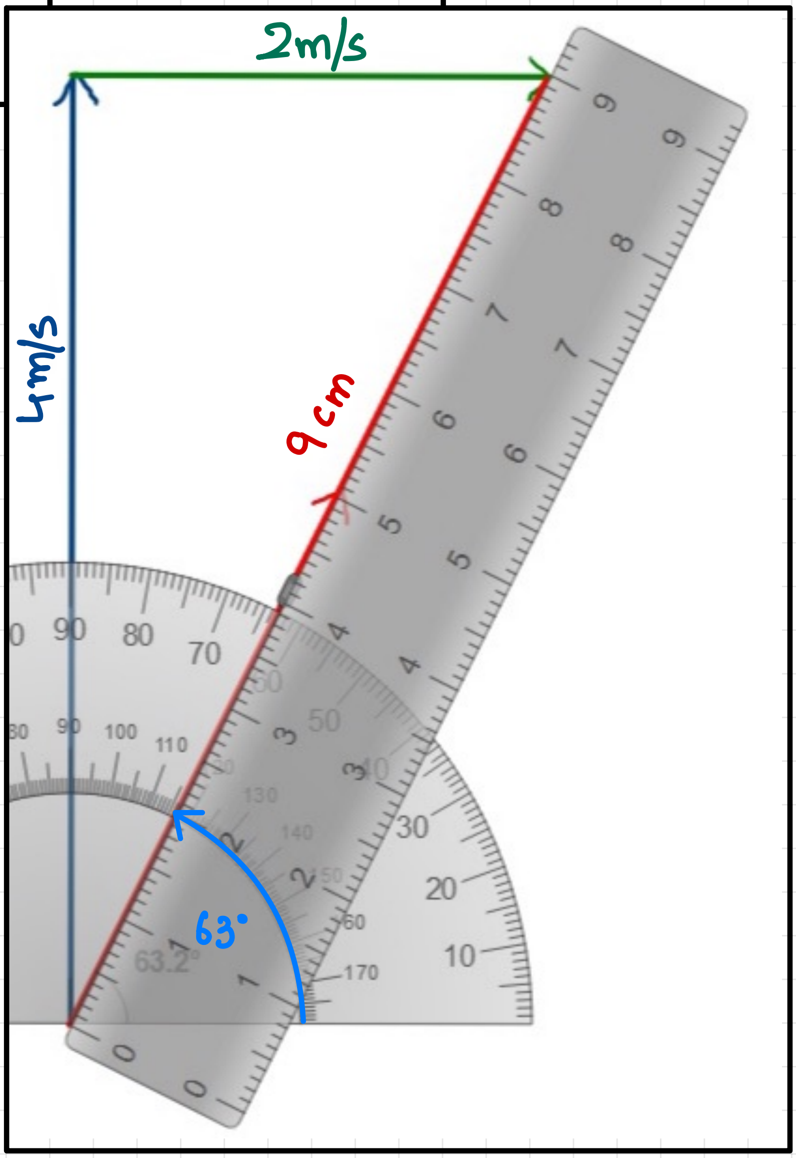
Scale

$1cm = 0.5 \frac{m}{s}$

Visualize representation of step 2



Visualize representation of step 3



- **Resultant's Magnitude:**

$$1cm = 0.5\frac{m}{s}$$
$$9cm = R$$

$$R = 9 \times 0.5$$

$$\boxed{R = 4.5\ \frac{m}{s}}$$

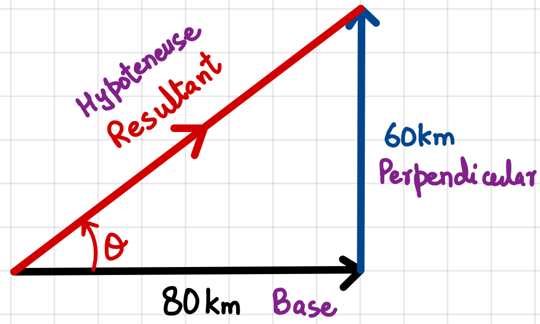
- **Resultant's Direction:**

63° with x-axis.

Right Angled Triangle:

Find out the magnitude of a resultant.

Base is from where the angle is formed.
Hypoteneuse is opposite to the 90° angle.
Perpendicular is the remaining side.



Pythagorus Theorem is used to find the magnitude of resultant vector.

$$H^2 = B^2 + P^2$$

$$\sqrt{R^{\cancel{2}}} = \sqrt{80^2 + 60^2}$$

$$R = \sqrt{6400 + 3600}$$

$$R = \sqrt{10000}$$

$$\boxed{R = 100\text{ km}}$$

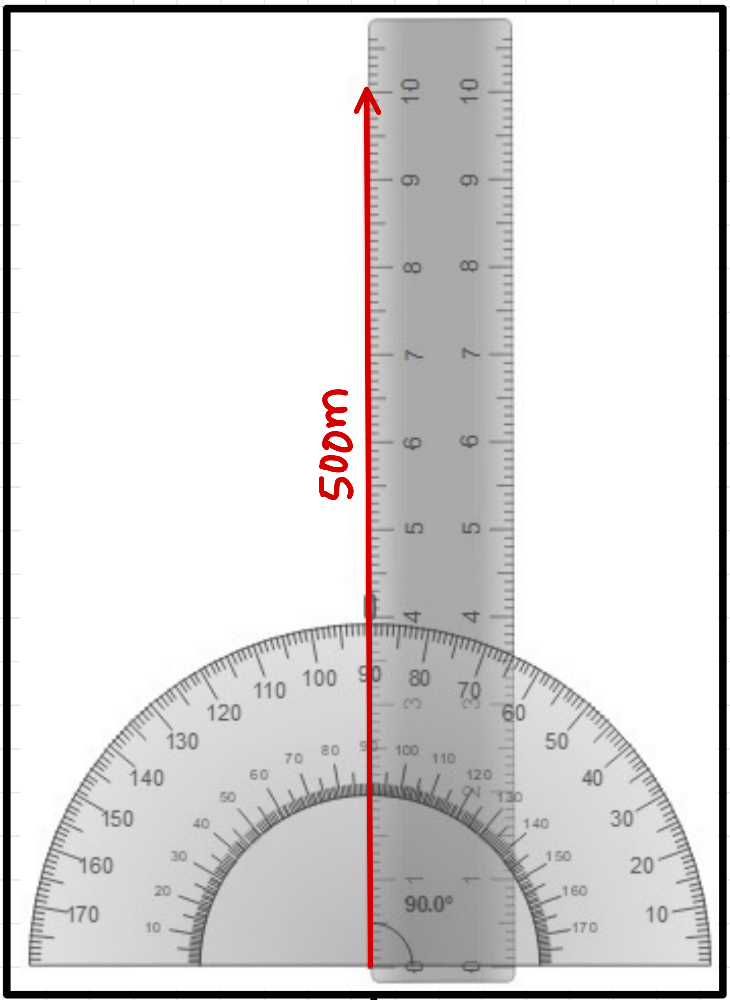
SCAN ME

Parallelogram Method on Youtube

Parallelogram Method:

1. Choose 1st vector i.e. 500m towards north. Draw this vector according to scale.
2. Draw 2ND vector of 200m towards east by joining the tail of 2ND vector with the tip of 1ST vector without any change in magnitude & direction.
3. Draw parallel vector of 2ND vector in front of it without any change in magnitude & direction.
4. Draw parallel vector of 1ST vector in front of it without any change in magnitude & direction.
5. Draw resultant vector which is the diagonal of parallelogram which starts from both tails & ends at both tips. Represent it with double arrows.
6. Measure the length of resultant vector & convert it according to scale. This is the magnitude of resultant vector.

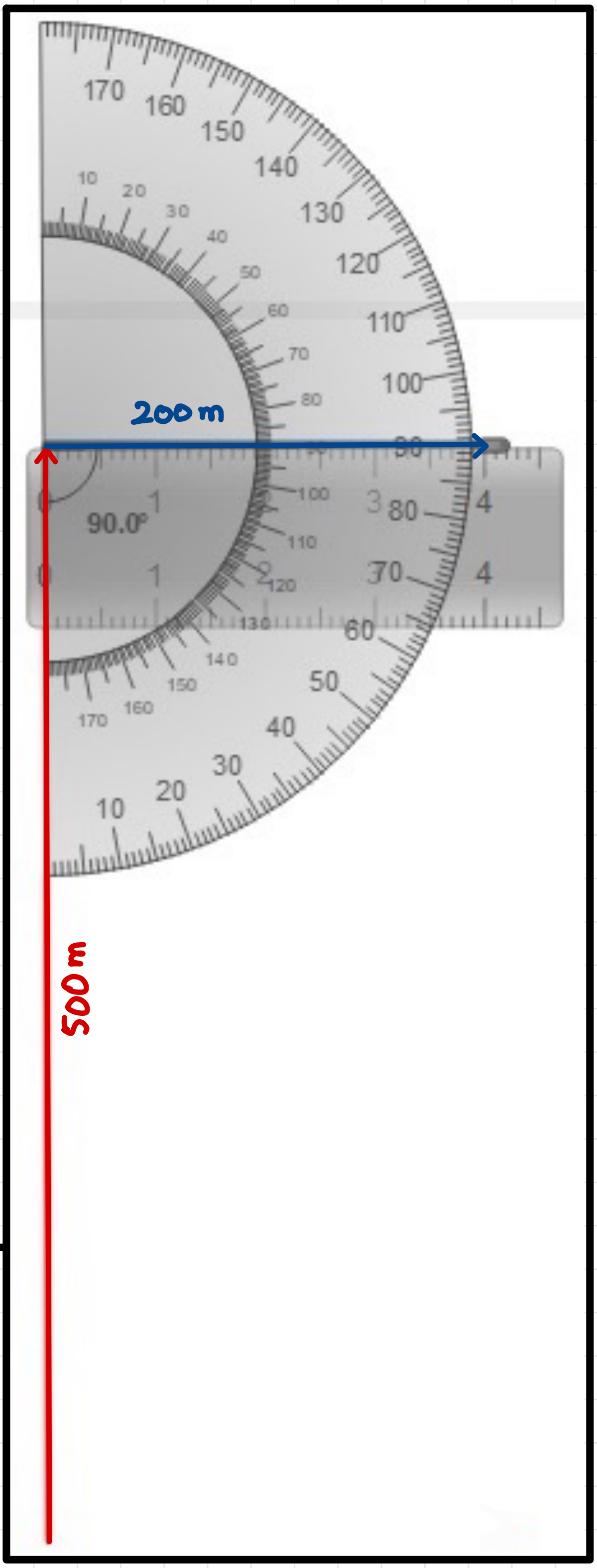
⑦ Measure the angle of resultant vector. This is the direction of resultant vector.



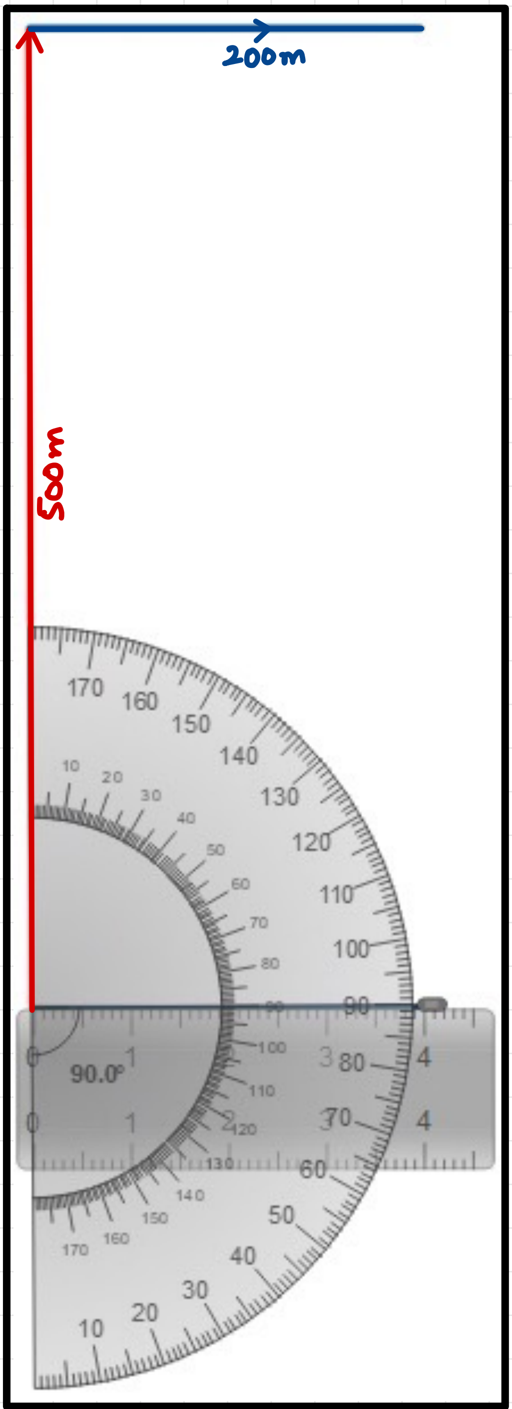
Scale

1cm = 50m

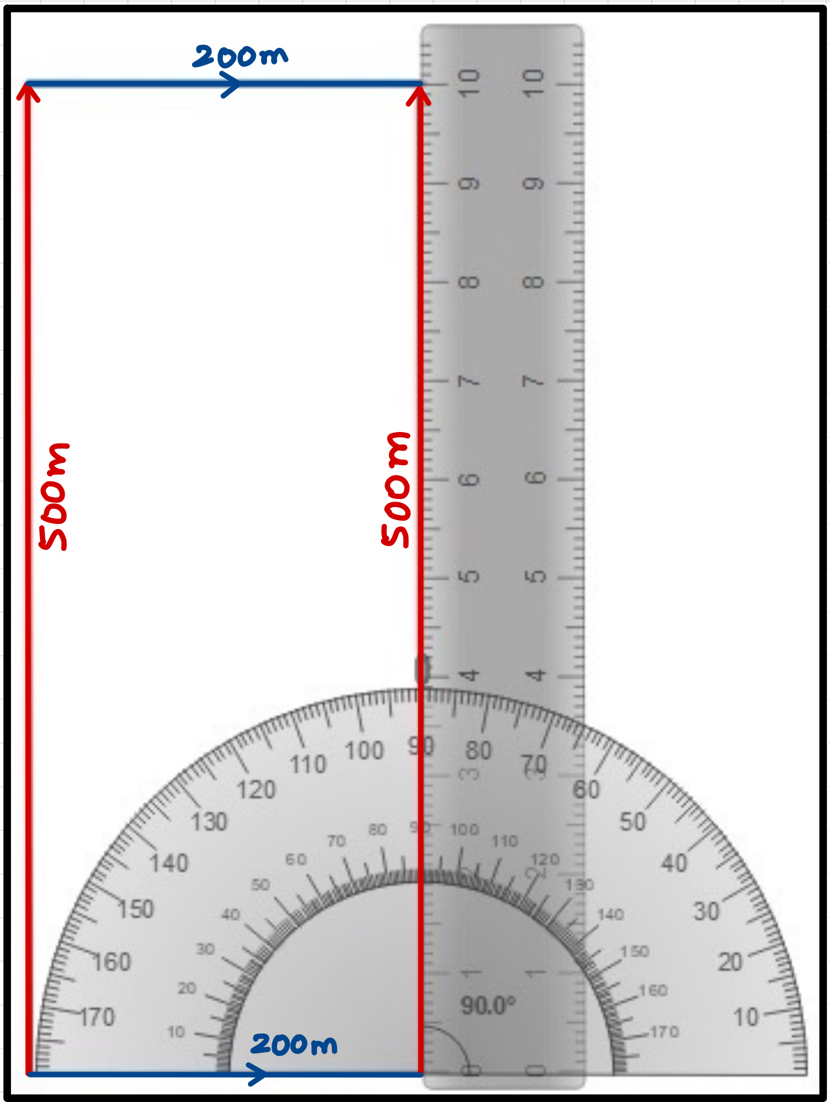
Visualize representation of step 1



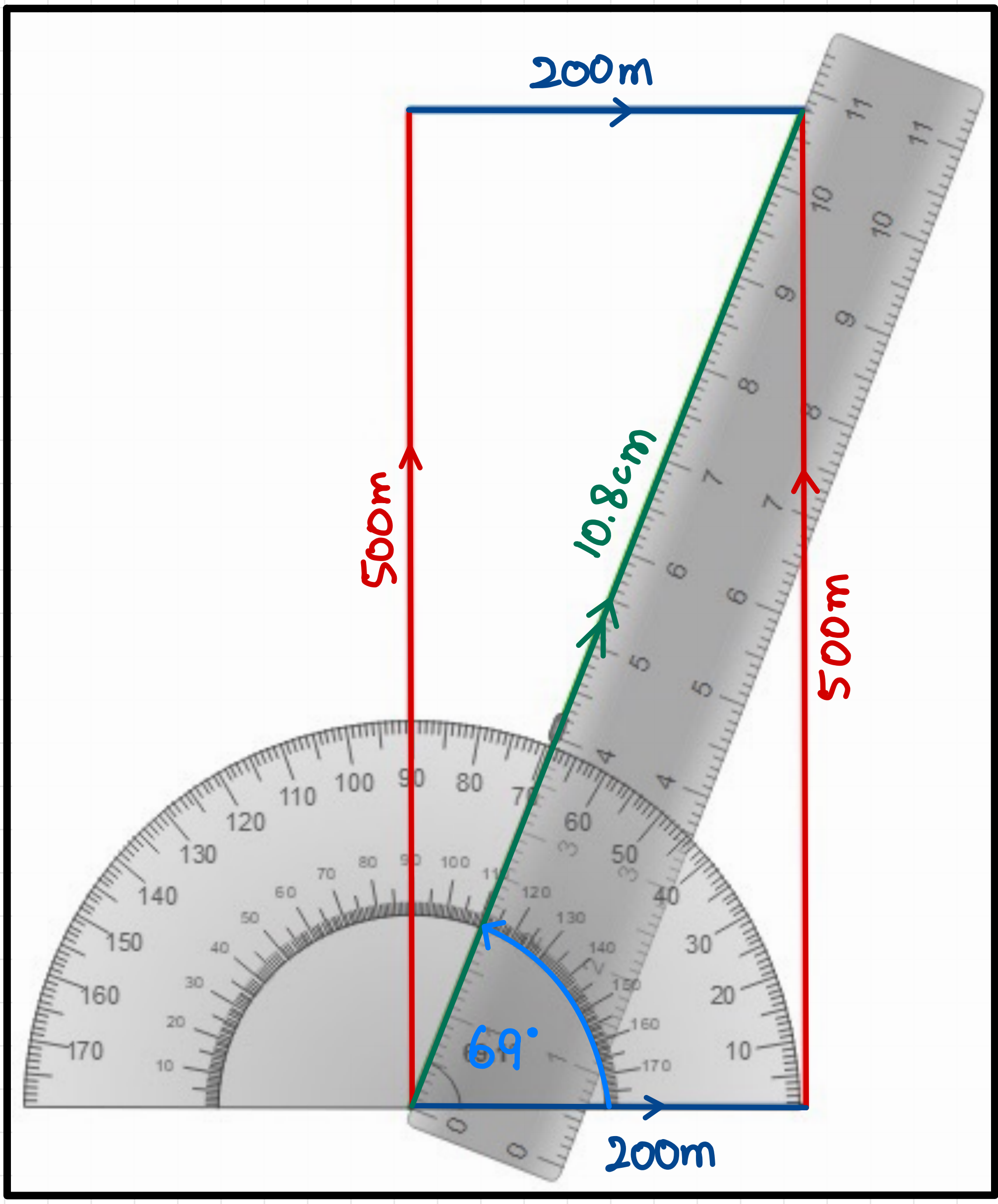
Visualize representation of step 2



Visualize representation of step 3



Visualize representation of step 4



Visualize representation of step 5

- **Resultant's Magnitude:**

1cm = 50 m
10.8 cm = R

$\Rightarrow$ $R = 10.8 \times 50$ $\Rightarrow$ $\boxed{R = 540\ m}$

- **Resultant's Direction:**

69° with x-axis.

MATCH THE FOLLOWING

1 ms =	measured by micrometer screw gauge.
Less than 2cm length is	10^{-3} s
Time period defines	which arises due to incorrect line of sight.
Parallax error is the error	Time required to complete one oscillation.
Human reaction error	keep your eye perpendicular to the reading.
To avoid parallax error,	can be minimized by taking multiple reading & an average.
Scalars have magnitude	as well as specific direction.
Vectors have magnitude	but no specific direction

1 A plumber measures, as **accurately** as possible, the length and internal diameter of a straight copper pipe.

The length is approximately 80 cm and the internal diameter is approximately 2 cm.

What is the best combination of instruments for the plumber to use?

	internal diameter	length
A	rule	rule
B	rule	tape
C	vernier calipers	rule
D	vernier calipers	tape

[MJ2011/P11/Q1]

2 What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg/m^3
D	weight	kg

[MJ2011/P11/Q2]

3 The diameter and the length of a thin wire, approximately 1 m in length, are measured as accurately as possible.

What are the best instruments to use?

	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

[ON2011/P11/Q1]

4 A quantity is quoted as having a value of 6.2 ms.

In what units is it measured?

A metres

B metres per second

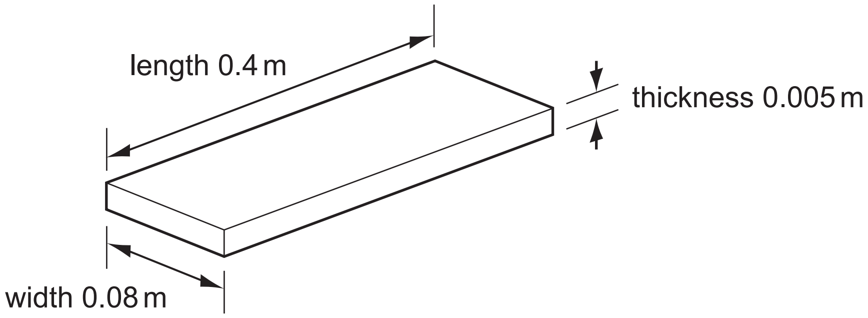
C microseconds

D milliseconds

[ON2011/P11/Q2]

5 A manufacturer measures accurately the dimensions of a wooden floor tile.

The approximate dimensions of the tile are shown.



Which instruments are used to measure accurately each of these dimensions?

	length	thickness	width
A	metre rule	micrometer	vernier calipers
B	metre rule	vernier calipers	micrometer
C	micrometer	metre rule	vernier calipers
D	vernier calipers	micrometer	metre rule

[MJ2012/P11/Q1]

6 Which pair of quantities includes one scalar and one vector?

A mass time

B temperature time

C temperature velocity

D velocity weight

[MJ2012/P11/Q2]

7 A reel of copper wire is labelled ‘length 30 m’ and ‘diameter 2 mm’. A student calculates the volume of the copper wire.

Which instruments does he use to measure accurately the length and the diameter of the wire?

	length	diameter
A	rule	calipers
B	rule	micrometer
C	tape	calipers
D	tape	micrometer

[MJ2012/P12/Q1]

8 Which row correctly shows examples of a vector quantity and a scalar quantity?

	vector	scalar
A	area	force
B	mass	density
C	velocity	acceleration
D	weight	volume

[MJ2012/P12/Q2]

9 Velocity is given by the change in displacement divided by the change in time.

How many vector quantities appear in this statement?

A 0 **B** 1 **C** 2 **D** 3 **[ON2012/P11/Q2]**

10 The level of water in a measuring cylinder is $75\,cm^3$. A stone of volume $20\,cm^3$ is lowered into the water.

What is the new reading of the water level?

A $20\,cm^3$ **B** $55\,cm^3$ **C** $75\,cm^3$ **D** $95\,cm^3$ **[ON2012/P12/Q1]**

11 The diagram shows three forces acting on a block.

Which additional force will produce a resultant force of 3 N to the left?

A 3 N to the left

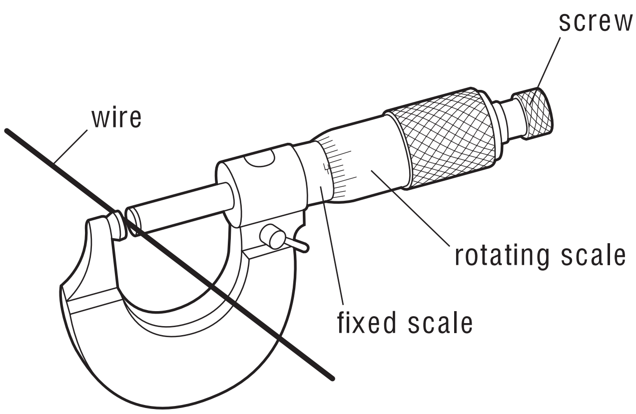
B 6 N to the right

C 9 N to the left

D 13 N to the right

[ON2012/P12/Q2]

12 A micrometer is used to measure the diameter of a uniform wire.



What is done to obtain an accurate answer?

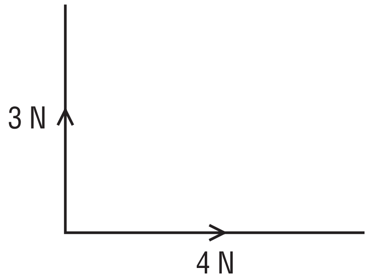
A Find the reading and add or subtract the zero error.

B Make the micrometer horizontal.

C Subtract the fixed scale reading from the rotating scale reading.

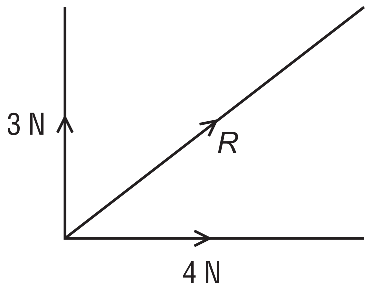
D Subtract the rotating scale reading from the fixed scale reading. **[MJ2013/P11/Q2]**

13 Forces of 3 N and 4 N act as shown in the diagram.

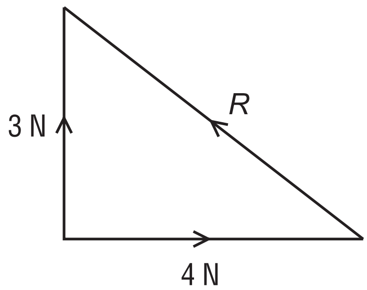


Which diagram shows the resultant *R* of these two forces?

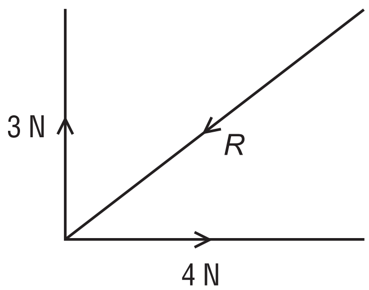
A



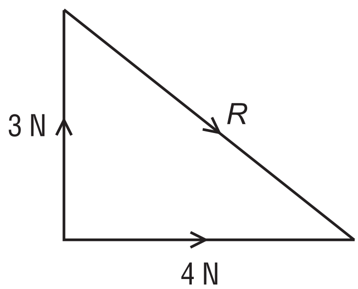
B



C



D



[MJ2013/P11/Q1]

14 Before marking the finishing line on a running track, a groundsman measures out its 100 m length.

Which instrument is the most appropriate for this purpose?

A measuring tape

B metre rule

C 30 cm ruler

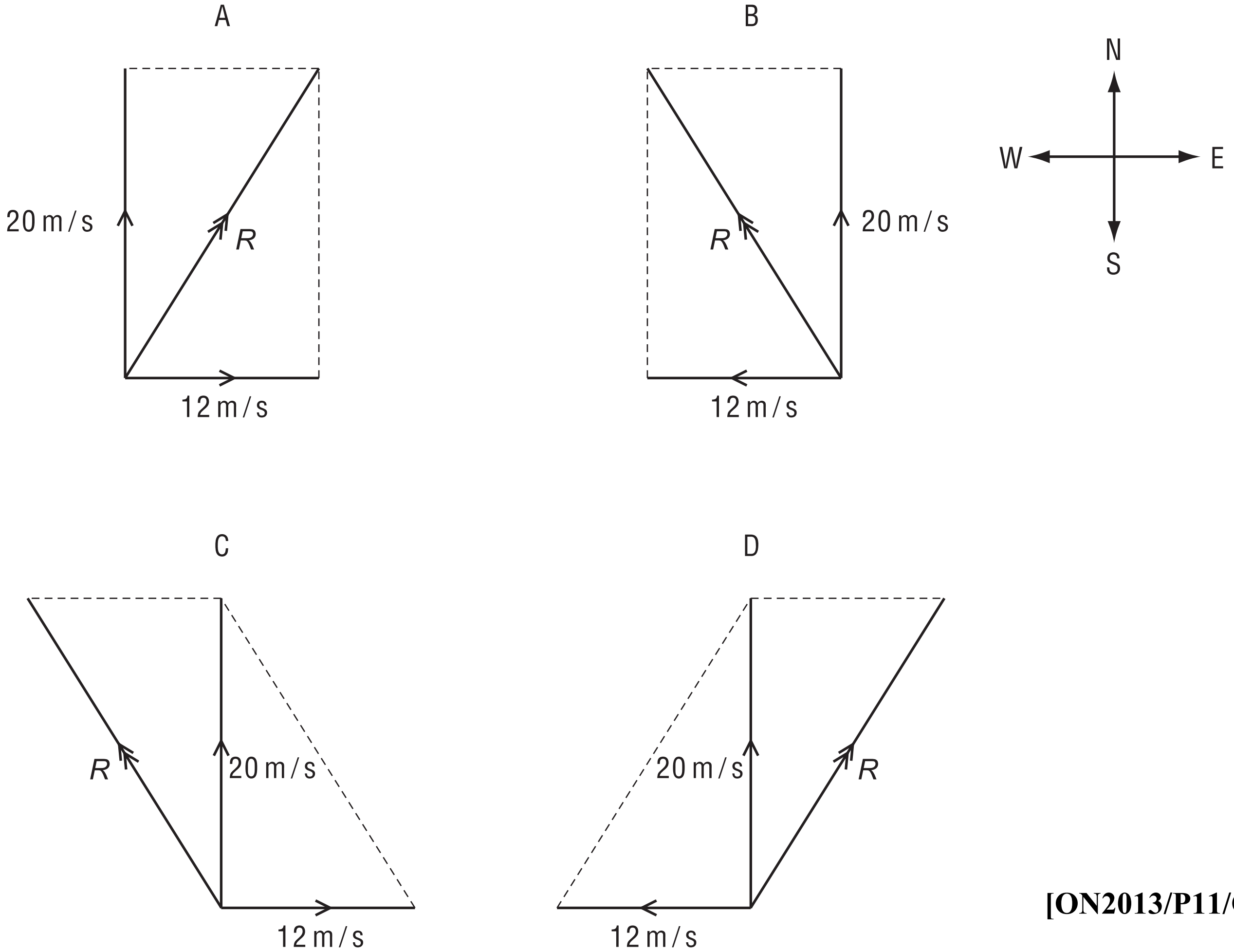
D micrometer

[MJ2013/P12/Q2]

15 When there is no wind, the engines of an airship push it due north at 20 m/s.

The wind is blowing from the west at 12 m/s.

Which vector diagram correctly shows how the resultant velocity *R* of the airship is obtained?



[ON2013/P11/Q1]

16 A length of copper pipe, of uniform cross-section and several metres long, carries water to a tap. Measurements are taken to determine accurately the volume of copper in the pipe.

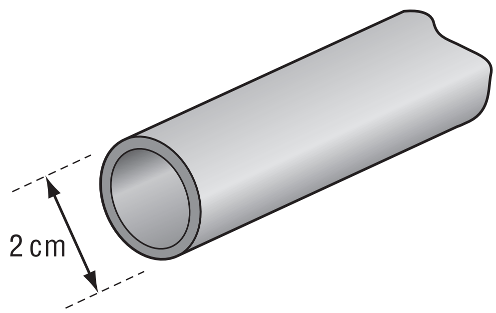
Which instruments are used?

A calipers and micrometer

B micrometer and rule

C rule and tape

D tape and calipers



[ON2013/P11/Q2]

17 A workman measures, as **accurately** as possible, the length and internal diameter of a straight copper pipe.

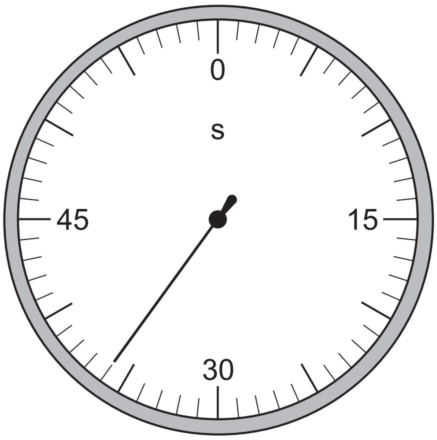
The length is approximately 600 cm and the internal diameter is approximately 2 cm.

What is the best combination of instruments for the workman to use?

	internal diameter	length
A	ruler	ruler
B	ruler	tape
C	vernier calipers	ruler
D	vernier calipers	tape

[MJ2014/P11/Q1]

18 The diagram shows a stopwatch.



What is the reading on the stopwatch?

A 30.6 s **B** 33.0 s **C** 36.0 s **D** 36.6 s

[MJ2014/P11/Q2]

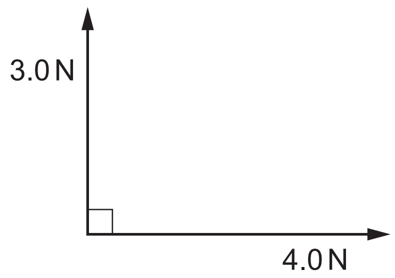
19 Each row contains a vector and a scalar.

In which row is the size of the vector equal to the size of the scalar?

	vector	scalar
A	displacement of a car	speed of the car
B	velocity of a car	distance travelled by the car
C	velocity of a car	speed of the car
D	weight of a car	mass of the car

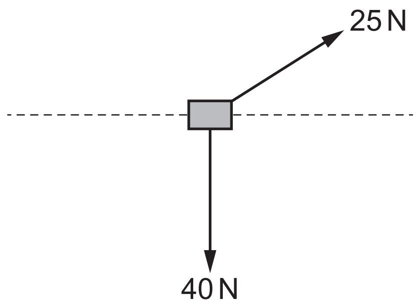
[MJ2014/P12/Q1]

20 What is the size of the resultant of the two forces shown in the diagram?

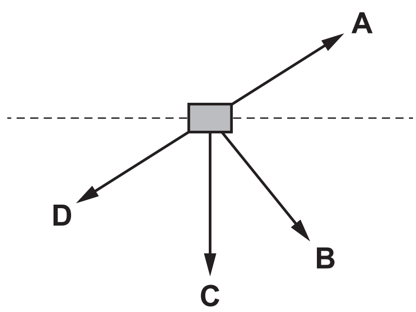


A 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N **[MJ2014/P12/Q2]**

21 Forces of 25 N and 40 N act on an object in the directions shown.



Which arrow shows the direction of the resultant force on the object?



[ON2014/P11/Q1]

22 Which device can be used to measure the thickness of a single sheet of paper?

A a metre rule

B a micrometer

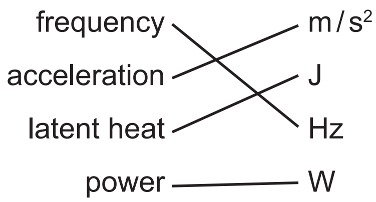
C a plastic ruler

D a measuring tape **[ON2014/P11/Q2]**

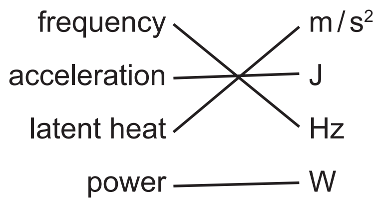
23 In a test, four students linked the quantities on the left with their units on the right.

Which student matched them all correctly?

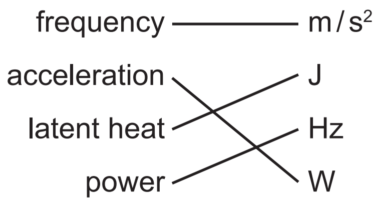
A



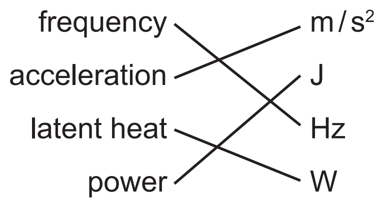
B



C



D



[ON2014/P11/Q3]

24 Is mass a scalar or a vector, and is acceleration a scalar or a vector?

	mass	acceleration
A	scalar	scalar
B	scalar	vector
C	vector	scalar
D	vector	vector

[MJ2015/P11/Q1]

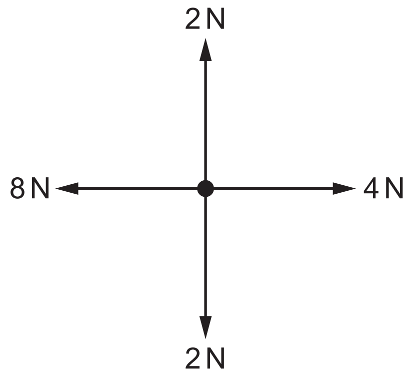
25 The diameter and the length of a thin wire, approximately 50 cm in length, are measured as precisely as possible.

What are the best instruments to use?

	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

[MJ2015/P11/Q2]

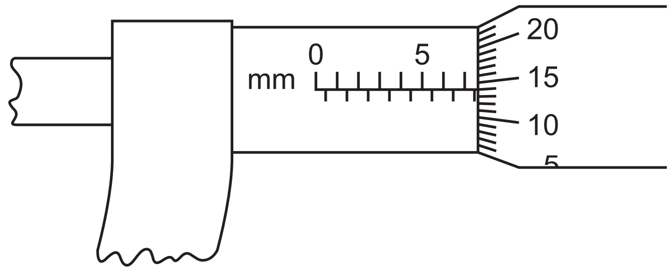
26 Four forces act at a point as shown.



What is the size of the resultant force?

A 0N **B** 4N **C** 6N **D** 8N **[MJ2015/P11/Q6]**

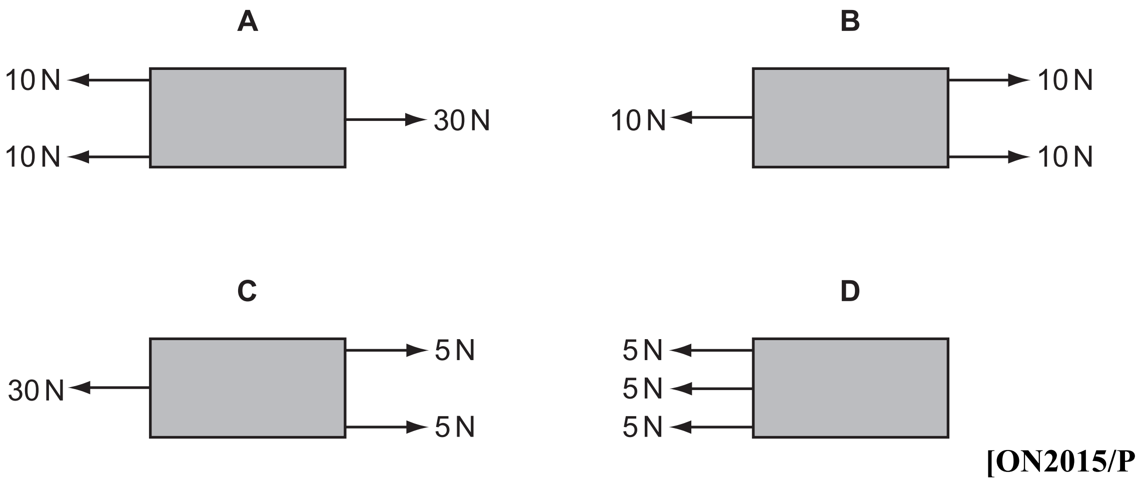
27 The diagram shows a micrometer scale.



Which reading is shown?

A 5.64mm **B** 7.14mm **C** 7.16mm **D** 7.64mm **[ON2015/P11/Q2]**

28 Which object has the largest resultant force?



[ON2015/P11/Q4]

ANSWER KEY			
1	C	21	B
2	B	22	B
3	A	23	A
4	D	24	B
5	A	25	A
6	C	26	B
7	D	27	D
8	D	28	C
9	C		
10	D		
11	C		
12	A		
13	A		
14	A		
15	A		
16	D		
17	D		
18	C		
19	C		
20	C		

1 A parachutist of total weight 950 N falls vertically at a constant velocity of 6.5 m/s.

① 6.5 m/s

4.5 m/s

(a) State the size and direction of the resistive (drag) force acting on the parachutist.

size of resistive force: ....

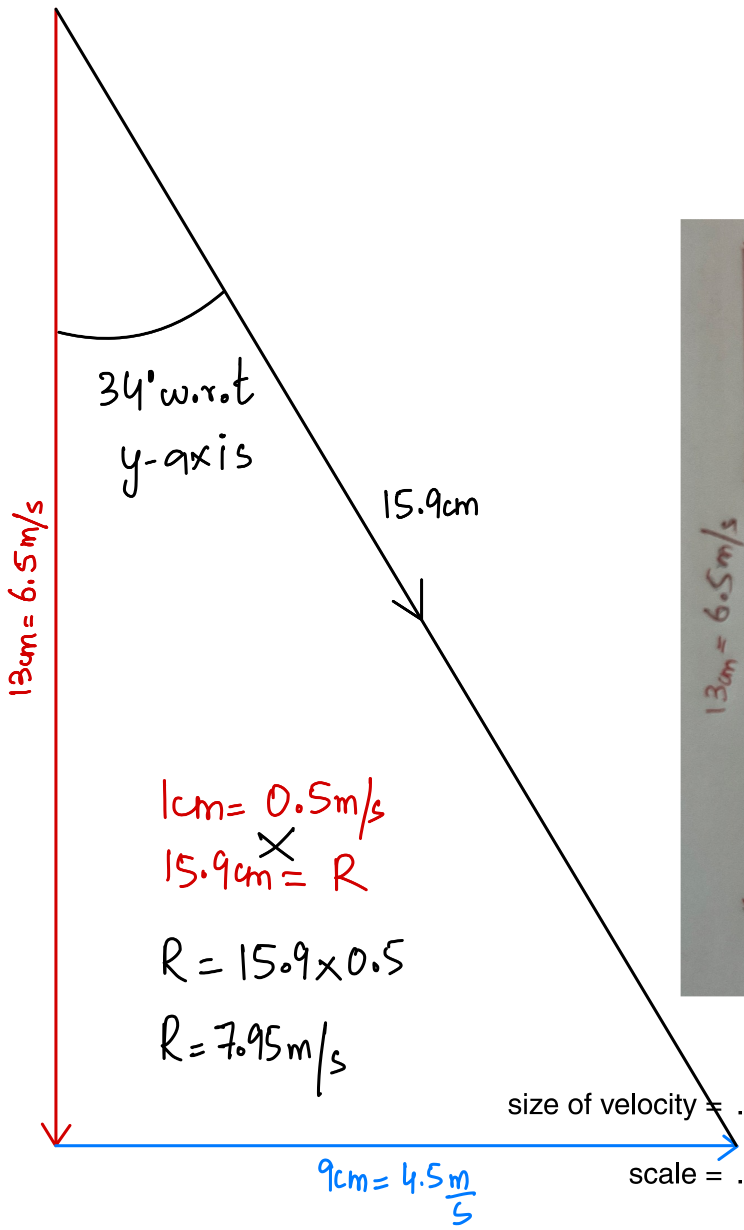
direction of resistive force: ....

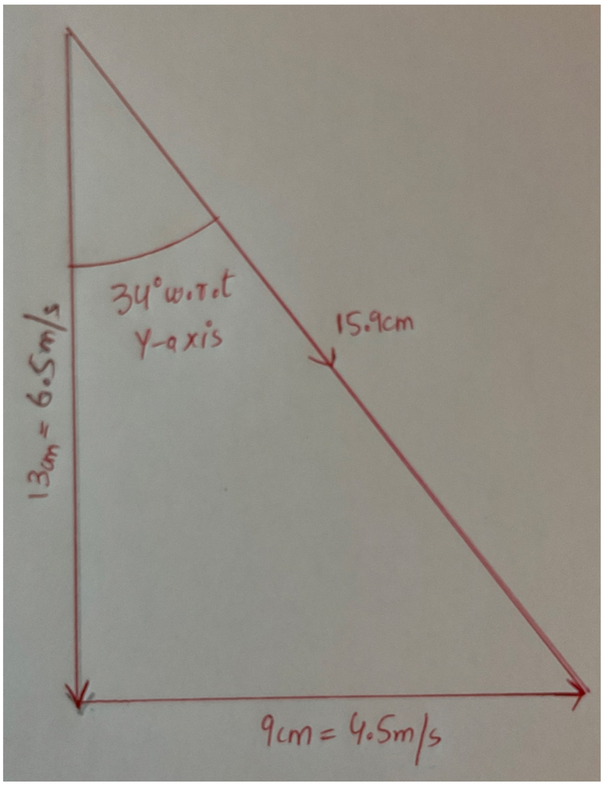
[2]

(b) At a certain height, the wind starts to blow and the parachutist moves horizontally at a velocity of 4.5 m/s as he continues to fall.

Use a graphical method to add together the horizontal and the vertical velocities to determine the size of the resultant velocity of the parachutist. Label the velocities and state the scale that you use.

① Scale
② Diagram
③ Resultant Magnitude
④ Direction





size of velocity = ...7.95 m/s........................

scale = ...1cm = 0.5 m/s........................

9cm = 4.5 m/s

13cm = 6.5 m/s

[3]

[ON2012/P21/Q1]

2 A set of traffic lights hangs from the end of a metal cable. A horizontal chain pulls the traffic lights to the right so that they are above the middle of the road. Fig. 2.1 shows the metal cable inclined to the vertical.

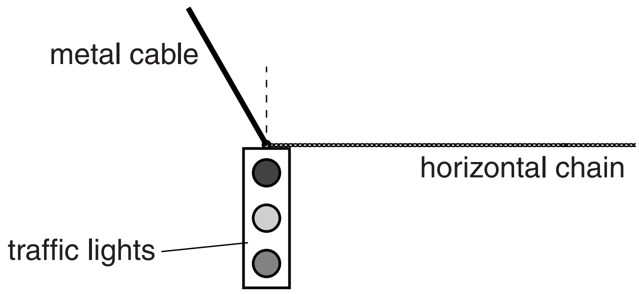


Fig. 2.1

The weight of the traffic lights is 240 N.

(a) Two of the forces on the traffic lights are the tension in the horizontal chain and the weight of the traffic lights.

On Fig. 2.1, mark

(i) an arrow that represents the tension in the horizontal chain, [1]

(ii) an arrow that represents the weight of the traffic lights. [1]

(b) The tension in the horizontal chain is 140 N. Use a scale diagram to determine the size of the resultant of the weight and the tension in the chain. State the scale used for the diagram.

scale = ...

resultant force = ...[3]

[ON2013/P22/Q1]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2012	5054	21

1	(a)	950 N	B1	
		upwards	B1	
	(b)	correct rectangle **and** diagonal **and** at least one velocity labelled **or** correct triangle and at least one velocity labelled (either way round)	B1	
		from 7.8(0000) to 8.0(0000) m / s (inclusive)	B1	
		scale stated	B1	[5]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2013	5054	22

1	(a)	(i)	arrow(head) on chain pointing to the right	B1	
		(ii)	vertical arrow downwards and part of arrow touching **or** within rectangle of lights **or** direction of arrow in **(i) and (ii)** correct (by eye)	B1	
	(b)		scale given (must have unit of cm:N **or** cm/N **or** N:cm **or** N/cm)	B1	
			correct triangle **or** rectangle (might be implied) and correct resultant (compulsory e.c.f. from **(i) or (ii)**: i.e. correct diagonal according to candidate's **(i)** and **(ii)**)	B1	
			272 ≤ candidate's value ≤ 283 N	B1	[5]